

Sahil Dangol

Full-Stack Gen AI Engineer

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PROFESSIONAL SUMMARY

Full-Stack Gen AI Engineer with **2 Years** of Experience specializing in **zero-knowledge RAG pipelines**, Agentic Workflows and **high-concurrency B2B financial systems**. Demonstrated record of shipping **8+ enterprise-grade projects** and **50+ technical projects**. Expert in bridging complex machine learning research and implementations (TFT, Quantized LLMs) with production-ready, containerized microservices. **NASA Space Apps Global Nominee** with specialized experience in **Hardware-Software Interfacing** via BLE/GATT and real-time state synchronization.

TECHNICAL CORE

- **GenAI & LLM Infrastructure:** LangGraph,RAG/CAG Pipelines, Agentic Reasoning, LORA/QLORA Fine-Tuning, Quantization (GGUF/AWQ), LangChain,Local and Cloud Inference (Ollama/API Keys).
- **Quantitative ML & Research:** PyTorch, **Temporal Fusion Transformers (TFT)**, Scikit-learn,Gradient Boosting Algorithms, Neural Networks (RNN,CNN) , Feature Engineering, Multi-horizon Forecasting, Market Regime Filtering.
- **Backend & Mobile:** Python (FastAPI), Node.js (Nest.js), **Flutter (Dart)**, PostgreSQL, Redis, Docker, CI/CD, Kafka,RabbitMQ,MongoDB,Bluetooth Classic, **BLE/GATT Protocols**, Packet Encoding/Decoding, Azure Services.
- **Frontend :** React, Next.js, TypeScript, Vue.js,Tailwind CSS, Bootstrap,MUI,Framer Motion, Three.js.

PROFESSIONAL EXPERIENCE

Asteroid Studios	Lalitpur, Nepal
<i>Software Engineer</i>	May 2025 – Present

- **B2B FinTech Architecture:** Engineered core infrastructure for [books.eviebillings.com](#), delivering **10+ enterprise modules** including real-time FX integration and a dynamic **Custom Fields engine** for schema extensibility.
- **IoT / Mobile Development:** Architected and developed the **MWC Wireless Earbuds App** (Flutter), implementing **Bluetooth Classic** pairing and **BLE GATT**communication for hardware firmware control.
- **Hardware Interfacing:** Engineered custom **packet encoding/decoding** for a real-time Equalizer (EQ) feature in **MWC Earbuds Application**, maintaining precise hardware state synchronization with a sub-30MB app footprint.
- **Infrastructure Optimization:** Developed a containerized PDF reporting pipeline, reducing rendering latency by **40%** and maintaining **99.9% uptime** for high-concurrency transaction workflows.
- **Team Leadership:** Accelerated feature delivery by **25%** through modular frontend architecture and Served as Associate Project Manager on Headless CMS Project.

KathFOSS (Open Source Community)	Kathmandu, Nepal
<i>President</i>	Jan 2025 – Present

- Leading a national community of **100+ active contributors**, managing open-source project lifecycles using **Agile/Scrum** and enforcing strict production-grade CI/CD standards.

SELECTED ENGINEERING PROJECTS

AutoTFT Financial Forecasting Engine (Quantitative ML)

[\[GitHub\]](#)

- Implemented **Temporal Fusion Transformers (TFT)** for multi-horizon return prediction in the Nepali banking sector, outperforming standard XGBoost baselines in time-series accuracy.

Agentic AI Therapist (Privacy-First GenAI)

[\[GitHub\]](#) | [\[Demo\]](#)

- Developed a zero-knowledge AI system using **Cloud inference (Gemini API Key)** and a hybrid RAG/CAG architecture with PII to maximize data security.

Titan City: 3D Titan Exploration Game (Unity)

NASA Space Apps 2023 [\[Demo\]](#)

- Developed a complex Unity-based Exploration Game with custom physics and resource-management mechanics. Awarded **National Winner** and **NASA Space Apps Global Nominee**.

Quantitative NEPSE Stock Predictor

[\[Demo\]](#)

- Built an XGBoost model with SMA-based market regime filtering to forecast short-term equity momentum for the Nepal Stock Exchange.

PDF-RAG Brain (Mistral-7B Intelligence)

[\[Demo\]](#)

- Built an end-to-end semantic search engine utilizing PDF tokenization and Vector DB storage for highly accurate natural language retrieval.

EDUCATION & SPECIALIZATION

Tribhuvan University, Kathford Engineering College

2023 – 2027

Bachelor of Computer Engineering | Distinction (Top 5% of Cohort)

Rice University - Houston, Texas

2025

Engineering Project Management Specialization | Focused on technical risk assessment and lead-time optimization for complex systems.

AWARDS & RECOGNITION

- **NASA Space Apps Hackathon (2023): Global Nominee & National Winner.** Recognized for "Titan City" (Unity/C#).
- **Nationwide Hackathon (2024): Winner.** Engineered an AI-powered agricultural diagnostic assistant for rural farmers.